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Media Engineering and Technology
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CSEN 401 - Computer Programming Lab, *Spring 2024*

DoorDasH: Scare vs Laugh Touchdown

Milestone 3

Deadline: 18.5.2026 @ 11:59 PM

In this milestone, you are required to implement the GUI to be able to play the game.

1 General Guidelines

- The effects of any action performed in the GUI should be reflected in the engine and vice versa.
- The player should be able to view all content at all times without the need to resize/minimize/maximize the window during runtime.
- The action that is currently happening in the game should always be clearly indicated in the GUI.
- Make sure to handle all exceptions and validations for any input or action performed. In case any exception implemented in the second milestone arises the player should be notified and the action should be prohibited and another action should be chosen by the player.
- The game should not be stopped/ terminated for any exception thrown. However, clicking the 'X' button on the window must be able to terminate the game at any instant.
- Try to adhere to the MVC architectural pattern to organize the codebase, enhancing maintainability and scalability:
 - **Model:** Manages the data, logic, and rules of the application independently of the user interface.
 - **View:** Represents the GUI which displays game information to the player. The view should be dynamic and reflect changes made in the model.

- **Controller:** Accepts input, converting it into commands for the model or view. This ensures that the UI is separated from the data processing.
- You are **not allowed** to use Swing for the GUI, only JavaFX is allowed.
- You're free to use SceneBuilder when building the GUI.

2 GUI Requirements

The requirements that should be covered in the GUI are explained below. You will be graded based on the requirements detailed in the following checklist:

Must be displayed whenever the player starts the game
A way to select a side (either SCARER or LAUGHER)
A way to start and play the game according to the chosen side
Game instructions
Game board with all 100 cells in their correct positions and types
Cell index number visible on each cell
Visible difference in cell types (SCARER doors, LAUGHER doors, Monster Cells, Card Cells, Conveyor Belts, Contamination Socks, Normal Cells)
Doors energy value must be visible
Both monsters initialized at cell 0 with their correct starting energy
25 shuffled cards for card cells
Player must be able to perform the following actions
Choose whether to activate their powerup before rolling
Roll the dice to move their monster
Distinguish between monster types
Player should keep track of the following throughout the game
Current turn
Current player and opponent
Result of each dice roll
Card drawn whenever a Card Cell is landed on, including its name and effect
Player skipping their turn (Freeze effect) must be clearly indicated
Must be displayed for each monster
Name
Original Role (SCARER / LAUGHER)
Current Role to indicate confusion
Type (Dasher / Dynamo / Multitasker / Schemer)
Current energy
Current position on the board
Active status effects (shield, confusion turns remaining, etc)

Must be shown and updated on the board
Cell effects must be indicated (changed position, energy, etc)
Activated / exhausted door cells must be distinguishable from non-activated door cells
Monster Cells must display the stationed monster's identity
Any energy change to any monster must be indicated
Shield blocking an energy loss must be indicated
Must be shown and updated for each card
Card name and effect must be displayed when drawn
Card effect must be indicated (Swap, start over, etc)
Must be shown and updated whenever a player ends a turn
Both monsters' positions updated on the board
Energy values updated for all affected monsters
Status effect durations (Confusion, Momentum Rush, Focus Mode, Freeze, etc)
Role confusion must be visually indicated when a monster's role is temporarily swapped
Must be shown for any invalid action
Indicator to the player and reasons why the action could not be performed
The game should not be stopped / terminated for any invalid action
Closing the popup should NOT terminate the game
Must be shown whenever a player wins
Game Won / Game Over Screen
The winning monster's name and role must be announced
Final energy of both monsters must be displayed
Player must be able to return to the start window
Any specific one of the following can be done additionally
Fantastic GUI + Animations